Arpit Bhomia

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GitHub: github.com/KingAB2004 **Portfolio**: portfolio-arpitbhomia.netlify.app

Education

Indian Institute of Technology (IIT) Bhilai Chhattisgarh, India
B.Tech in Computer Science & Engineering; **CGPA: 9.45/10.0** (Transcript) August 2023 – May 2027

Work Experience

Software Engineering Intern
Visa Inc. | [Certificate] May 2026 – July 2026

- Built and integrated an **Autonomous UI Testing Agent** (LangChain, Playwright) into the developer workflow, replacing manual QA scripting and accelerating release cycles by reducing test automation time by **80%**.
- Expanded test coverage by **3x** via dynamic user inputs and engineered a RAG-based layer that cut LLM token consumption by ∼**40%**; utilized multithreading to significantly accelerate the heavy document processing workflows.
- Engineered a scalable video-to-document ingestion pipeline to translate UI session recordings into automated test flows; deployed a human-in-the-loop validation dashboard that reduced **false-negative test failures by 90%**.

Achievements

ICPC Regionalist: Secured **AIR 40** at the **ICPC Kanpur Regional Contest** [Certificate]

IICPC Codefest Regionalist: Secured **Global Rank 341** (Top 3%) [Certificate].

IICPC Quantfest Regionalist: Secured **Global Rank 286** (Top 2%) [Certificate].

Expert (Max Rating: **1810**, Top 3.95%) on Codeforces [Profile] and **5-Star Coder** (Max Rating: **2007**) on CodeChef [Profile]

Solved **450+ problems** on LeetCode [Profile] and **2000+ problems** overall across platforms.

Projects

TermiChat | [GitHub] | [Video] July 2025 – August 2025

- Architected a decentralized **P2P** terminal chat supporting AES-256 encrypted messaging, secure file transfers, and permission-based handshakes using custom raw **TCP** sockets.
- Optimized concurrency via a thread-safe producer-consumer queue and mutex locks, achieving ∼**150** msgs/sec throughput while synchronously persisting chat history to **SQLite3**.
- **Stack:** C/C++, POSIX Sockets, OpenSSL, SQLite3, Multithreading, ncurses

Warranty Storage Platform | [GitHub] | [Website] August 2024 – January 2025

- Engineered a scalable **MERN** warranty management platform with secure JWT authentication and **10MB** media processing, managing lifecycle telemetry for 500+ active enterprise records.
- Integrated a **LLaMA-3 RAG AI Chatbot** , reducing MongoDB database load by 70% via optimized client-side data caching and improving automated query resolution.
- **Stack:** MongoDB, Express.js, React.js, Node.js, Groq LLaMA-3, SendGrid API, JWT

CP Butler | [GitHub] | [Video] May 2025 – June 2025

- Developed a **Chrome extension** leveraging external REST APIs to consolidate user statistics and monitor contest schedules, reducing cross-platform tracking time by over **80**% via a centralized dashboard.
- **Stack:** JavaScript, HTML, CSS, Chrome Extensions API, REST APIs

Technical Skills

Programming Languages: C/C++, Python, JavaScript, TypeScript
Frameworks & Libraries: React.js, Node.js, LangChain, Playwright, TensorFlow, Crow
Databases: MongoDB, MySQL, SQLite
Relevant Coursework: Linux, Git/Github, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management System

Leadership & Activities

Coordinator – Ingenuity Competitive Programming Club, IIT Bhilai | [Certificate] April 2025 – March. 2026

- Led weekly sessions and 6+ contests for 100+ students, yielding **4 ICPC regional qualifiers**.